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Quantum Machine Learning for Earth Observation: A Review and Future Prospects

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***Abstract*—Quantum Machine Learning (QML) is an emerging interdisciplinary field that leverages quantum computing principles to address complex computational challenges. Traditional machine learning (ML) and deep learning (DL) models face major challenges in Earth Observation (EO) due to the increasing volume and complexity of data. These challenges include computational intensity, energy consumption, and data management constraints. QML emerges as a promising paradigm for overcoming these barriers by harnessing quantum phenomena such as superposition, entanglement, and interference. This review aims at offering a detailed analysis of the state-of-the-art and new trends in QML application to EO. We methodically examine important advancements in the field of QML4EO, by exploring the fundamental ideas of QML. Our review methodology involved systematic searches across prominent scientific databases, based on our extensive knowledge on the topic, using carefully formulated queries to ensure broad coverage and high relevance. In addition, we address the changing institutional and geographic environment of QML4EO research, highlighting important centers of contribution and the important role of programs such as the IEEE GRSS QUEST Technical Committee. In order to transform QML from a specialized research area into a complementary and essential tool for advancing EO, this paper critically analyzes current limitations and provides an outlook on future directions, highlighting the need for reliable hardware, improved algorithmic design, and standardized protocols.**

***Index Terms*—Quantum Computing, Quantum Machine Learning, Remote Sensing, Earth Observation**

I. Introduction

The development of Artificial Intelligence (AI) has profoundly affected Earth Observation (EO) with its ability to extract meaningful information from a growing amount of data [1]. However, the integration of big data with AI-based EO is not without challenges. Despite advances brought forward by algorithms Machine Learning (ML) and especially Deep Learning (DL) in analyzing data EO, large-scale ML can be time-consuming and energy-intensive due to inherent limitations related to data management, processing, and effective resource utilization [2]. These limitations can be mitigated by improving computational capability and efficiency. Quantum Computing (QC) has emerged as a pioneering solution that leverages principles of quantum physics such as superposition, entanglement and interference to perform calculations [3].

Interest in implementing ML methodologies based on QC led to the creation of a new research field known as Quantum Machine Learning (QML) [4], [5]. Initial studies show that quantum-enhanced models can potentially explore larger solution spaces, leading to faster convergence, better generalization, and improved performance, especially for large and complex datasets [6].

The question of whether QC technologies and methods are compatible with EO tasks naturally arises. Interest in the field has been growing, with research primarily focusing on hybrid quantum-classical model architectures. This is motivated by the current capabilities of quantum hardware, which are limited in size and coherence time. Although significant, these advances only partially address the broader challenges and opportunities in deploying quantum-enhanced models in EO [7].

This research direction, summarized as Quantum Machine Learning for Earth Observation (QML4EO), has only emerged significantly in 2017 and is therefore relatively new. The total number of contributions is low, but has increased markedly in the last two years, as shown in Fig. 1.

This phenomenon is partially motivated by the fact that we are seeing a similar sharp increase in pure QC research, suggesting that significant breakthroughs may be close, and that we may be currently entering the «Quantum computers' "ChatGPT moment"» [1]. In contrast to more established research subjects that were yielding progressive results, there is a noticeable recent spike in research across various EO subfields, indicating a growing interest within the community towards this innovative and trans-

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[1] «Quantum computers' "ChatGPT moment"» https://www.bcg.com/capabilities/digital-technology-data/emerging-technologies/expert-insights/jean-francois-bobier

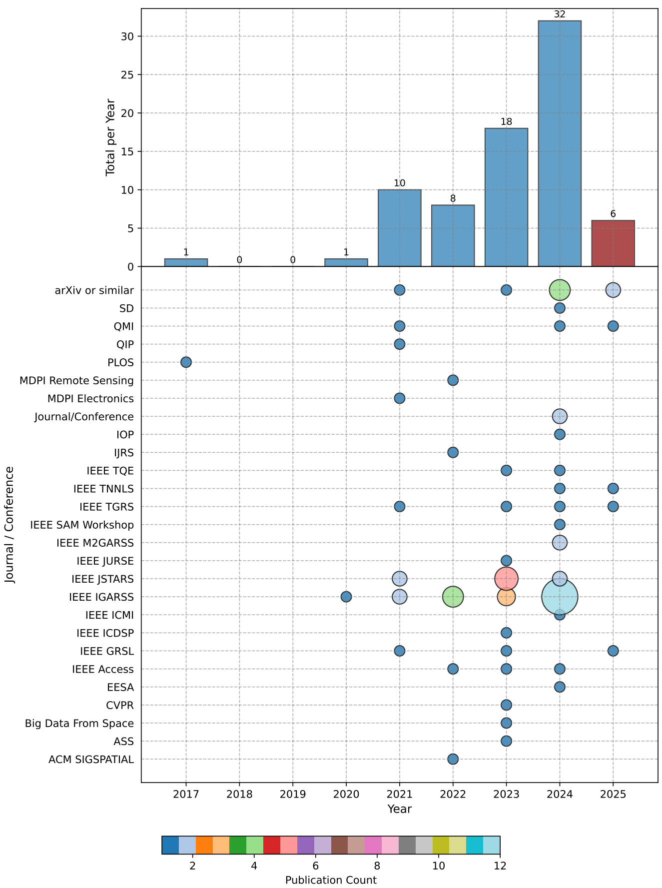

Figure 1. Number of papers on QML4EO over the last 8 years, distributed over different journals. Please note that data for 2025 represent an estimate. Nevertheless we are expecting a steady positive growth of publications in the field (e.g. post events like LPS2025 and IGARSS2025, papers submitted in 2025 and currentely on review).

formative computing paradigm. Beside its relative infancy, this field of research is motivated by the extraordinary potential impacts that this technology could have on EO.

In this paper, our aim is to engage a broader community by offering a less technical and more accessible perspective on this niche topic. Although existing reviews [8], [9] provide valuable information, they have limited scope and do not cover the latest advances in quantum machine learning for EO in a comprehensive way. Our work seeks to bridge this gap by presenting a broader, more inclusive, up-to-date overview, highlighting both opportunities and challenges. By doing so, we hope to foster interdisciplinary collaboration and drive further research in this emerging field.

The paper is structured as follows: i)

- In Sect. 2, the fundamental aspects of quantum information and QML are introduced.
- In Sect. 3, an exhaustive summary of the QML4EO literature is included, highlighting the chosen review methodology and the different classes of methods.
- In Sect. 4, information on application domains, trends, research ecosystems is extrapolated from the QML4EO literature.
- In Sect. 5, the presented status of the field is discussed, highlighting potential and limitations of QML.
- In Sect. 6, general conclusions are drawn on the review, and future prospects are proposed.

II. QML: Foundations and Key Concepts

QML lies at the intersection of QC and classical ML, aiming to enhance algorithmic capabilities by leveraging quantum properties such as superposition, entanglement, and interference. These principles enable the encoding and manipulation of data in high-dimensional Hilbert spaces using quantum states, potentially offering exponential speedups or memory advantages for certain problems. With the emergence of Noisy Intermediate-Scale Quantum (NISQ) devices—intermediate-scale quantum processors that operate without full error correction—QML has gained traction as a near-term strategy for solving problems in optimization, pattern recognition, and time-sensitive signal analysis. Preskill et al. [10] highlights how such devices may soon outperform classical computers in specific tasks, despite their inherent noise, and identifies hybrid quantum-classical algorithms (e.g, Quantum Approximate Optimization Algorithm (QAOA), Variational Quantum Eigensolver (VQE)) as particularly promising for early applications. In this context, QML becomes especially relevant for EO, where quantum-enhanced models could improve the classification of hyperspectral imagery, tracking of dynamic climate variables, and modeling of complex geophysical processes [11], [12].

A. Quantum Information Principles

At the heart of quantum computing lies the quantum bit (qubit), a two-level quantum system that generalizes the classical bit. A qubit exists in a coherent superposition of states $|0\rangle$ and $|1\rangle$:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \quad \text{where } |\alpha|^2 + |\beta|^2 = 1. \tag{1}$$

This state resides on the surface of the Bloch sphere, a geometrical representation where each point encodes a possible pure state. The manipulation of qubits is governed by unitary operations, preserving the norm of the state vector.

Another key concept is entanglement, i.e., a quantum correlation between multiple qubits that cannot be described classically. For two qubits, a maximally entangled Bell state is given by:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \tag{2}$$

Entanglement enables a form of quantum parallelism essential for QML algorithms, as it allows representing inter-dependencies among variables, for example, between temperature and pressure fields in atmospheric modeling.

Quantum circuits consist of sequences of quantum gates—unitary transformations applied to qubits—to implement computations. Gates such as the Hadamard, Pauli

(X, Y, Z), CNOT, Toffoli, and rotation gates form a universal set that can approximate any quantum operation [13]. Unlike classical logic gates, quantum gates are reversible and operate on complex-valued amplitudes. A common distinction is between single-qubit and multi-qubit gates.

Single-qubit gates (some examples in Tab. I) act on individual qubits and are represented by 2×2 unitary matrices. They manipulate the state by rotating it around the Bloch sphere. For example, the Hadamard gate (H) creates superposition by mapping basis states to equal-weight combinations. Rotation gates R_x, R_y, and R_z perform arbitrary-angle rotations around the respective axes, with θ specifying the rotation angle.

Table I
EXAMPLE OF SINGLE-QUBIT GATE OPERATIONS: X, Y, Z, H, AND ROTATIONS $R_x(\theta)$, $R_y(\theta)$, $R_z(\theta)$

Gate	Matrix	Description
H	$\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$	Creates equal superposition
$R_x(\theta)$	$\begin{bmatrix}\cos(\frac{\theta}{2}) & -i\sin(\frac{\theta}{2})\\ -i\sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2})\end{bmatrix}$	Rotation around x axis by angle θ
$R_y(\theta)$	$\begin{bmatrix}\cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2})\\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2})\end{bmatrix}$	Rotation around y axis by angle θ
$R_z(\theta)$	$\begin{bmatrix}e^{-i\theta/2} & 0\\ 0 & e^{i\theta/2}\end{bmatrix}$	Rotation around z axis by angle θ

Multi-qubit gates enable interactions between qubits and are essential for generating entanglement. The most common one is the controlled-NOT (CNOT) gate

$$CNOT = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\end{bmatrix} \tag{3}$$

CNOT is a two-qubit operation where the second (target) qubit is flipped if and only if the first (control) qubit is in state $|1\rangle$:

$$\text{CNOT}\,|00\rangle = |00\rangle\,, \quad \text{CNOT}\,|10\rangle = |11\rangle \tag{4}$$

These gates can be combined in sequence to build a quantum circuit where arbitrary unitary transformations are applied on multi-qubit systems. In fact, a finite set of gates, such as $\{R_x(\theta), R_z(\theta), \text{CNOT}\}$, forms a universal gate set capable of approximating any unitary operation to arbitrary precision. This sequence of gates that act on input states, modify the quantum state via interference, and extract classical information through measurement. Gate operations must be designed with circuit depth and coherence time constraints in mind, especially in the NISQ era.

Moreover, the notion of quantum measurement differs fundamentally from classical observation. Quantum measurements collapse the wavefunction into one of the eigenstates of the measured observable, making the process inherently probabilistic. Thus, output interpretation in QML involves repeated sampling of the same quantum circuit to estimate statistical properties (e.g., expectation values of observables).

Quantum information is also governed by no-cloning [14], [15] and no-deleting theorems [16], which set foundational constraints on data copying and erasure. These constraints impact the design of QML algorithms and limit redundancy mechanisms common in classical ML pipelines.

B. Data Encoding

A fundamental prerequisite for QML is the mapping of classical EO data into quantum states, a process known as *quantum data encoding*. This operation defines how information is represented in the Hilbert space and directly affects the model's expressive power, hardware efficiency, and resilience to noise.

Let $\mathbf{x} \in \mathbb{R}^n$ represent a classical input vector (e.g., a pixel spectrum or spatial feature descriptor). A quantum encoding routine prepares a quantum state $|\psi(\mathbf{x})\rangle$ such that the classical data is embedded into the amplitudes or parameters of the state. Formally:

$$|\psi(\mathbf{x})\rangle = U(\mathbf{x})\,|0\rangle^{\otimes n} \tag{5}$$

where $U(\mathbf{x})$ is a unitary transformation that encodes $\mathbf{x}$ into the quantum register. The main encoding scheme can be categorized as follow:

- **Angle Encoding:** Each feature x_i modulates a rotation gate $R_x(x_i)$ / $R_y(x_i)$ / $R_z(x_i)$ applied to the i-th qubit. This is hardware-efficient and suitable for near-term devices:
$$|\psi(\mathbf{x})\rangle = \bigotimes_{i=1}^{n} R_y(x_i)\,|0\rangle \tag{6}$$
For instance, Sebastianelli et al. [17] used this method to encode optical data.
- **Amplitude Encoding:** The vector $\mathbf{x}$ is normalized and used to directly specify the amplitudes of the quantum state:
$$|\psi(\mathbf{x})\rangle = \sum_{i=0}^{2^n-1} x_i\,|i\rangle\,, \quad \text{with } \sum_i |x_i|^2 = 1 \tag{7}$$
This allows exponential compression but requires complex state preparation, and it is typically less used when coming to hardware implementation.
- **Hybrid and Physics-Informed Embeddings:** EO-specific encoding schemes have been proposed to incorporate domain knowledge. For instance, Otgonbaatar et al. [18] introduced a method for encoding Stokes parameters from polarimetric SAR imagery into quantum gates, aligning the quantum representation with physical observables.

The choice of encoding impacts circuit depth, fidelity, and interpretability. For example, angle encoding is shallow and robust to noise, making it well-suited for NISQ

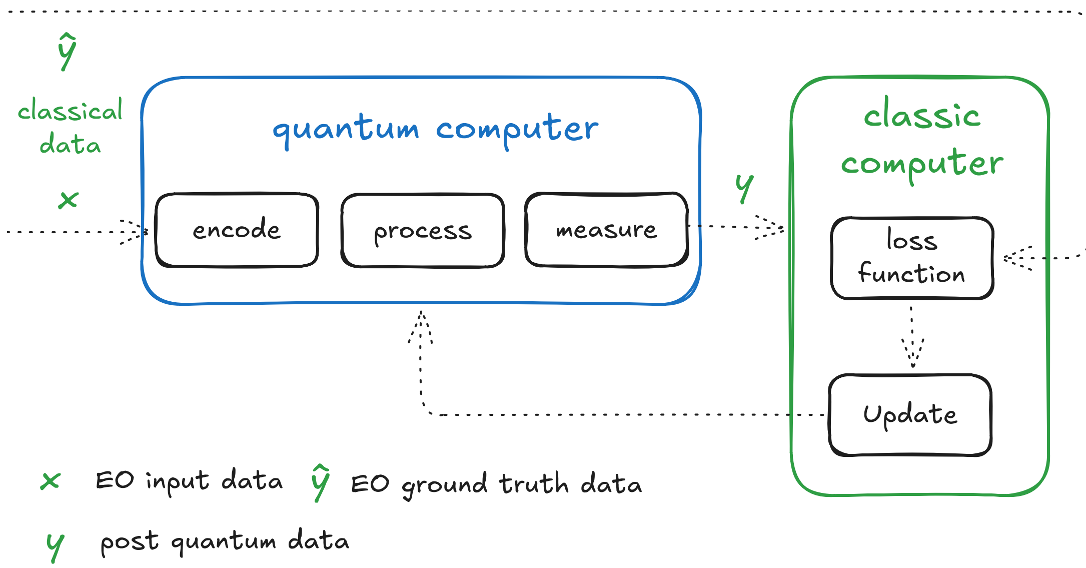

Figure 2. High level scheme representing the process of training a quantum machine learning model using a classical optimizer.

devices, while amplitude encoding offers higher expressiveness at the cost of deeper circuits and more complex state preparation [19].

C. Quantum Machine Learning Paradigms

QML includes a variety of algorithmic paradigms tailored to leverage QC for learning. Here we describe a few notable examples.

Variational Quantum Algorithms (VQAs) utilize Parameterized Quantum Circuits (PQCs) where parameters are iteratively optimized to minimize a classical objective function. This approach is well-suited to NISQ devices due to short circuit depth and hybrid computation. In EO, VQAs can be used for solving inverse problems such as deducing physical parameters from measurements:

$$\min_{\boldsymbol{\theta}} \langle \psi(\boldsymbol{\theta}) | \hat{H} | \psi(\boldsymbol{\theta}) \rangle \tag{8}$$

where $\hat{H}$ is an observable encoding the loss function, and $|\psi(\boldsymbol{\theta})\rangle$ is the quantum state.

Quantum Support Vector Machines (QSVMs) extend classical Support Vector Machine (SVM)s by defining quantum kernels that evaluate inner products in high-dimensional feature spaces. These kernels are computed via interference effects in quantum circuits and offer computational gains when classical kernels become costly, such as in multispectral pixel classification [11].

Quantum Principal Component Analysis (QPCA) enables efficient extraction of dominant eigenmodes from quantum-encoded density matrices. This is particularly relevant in dimensionality reduction of EO datasets, e.g., identifying major variance modes in sea surface temperature data using quantum phase estimation [11].

Quantum Neural Networks (QNNs) emulate neural architectures using quantum gates with tunable parameters. Circuit-based QNNs exploit superposition and entanglement to model complex functions and decision boundaries. For instance, they can be trained to detect abnormal vegetation indices or rainfall patterns from spectral time series [11].

D. Training Quantum Models

Training QML models requires efficient and hardware-aware optimization techniques (high level scheme presented in Fig. 2).

A widely used method for gradient-based learning is the *parameter-shift rule*, which provides exact gradients for expectation values under certain conditions:

$$\frac{\partial \langle \hat{O} \rangle}{\partial \theta} = \frac{1}{2} \left[\langle \hat{O} \rangle_{\theta + \frac{\pi}{2}} - \langle \hat{O} \rangle_{\theta - \frac{\pi}{2}} \right] \tag{9}$$

This method allows tuning parameters using classical optimizers and supports compatibility with the gradient descent and the Adam algorithm, widely used in deep learning [20].

Nevertheless, QML optimization landscapes are known to suffer from *barren plateaus*, where gradients vanish exponentially as the number of qubits or circuit depth increases. These regions make training difficult and require circuit designs that preserve local structure. Mitigation strategies include careful initialization (layerwise, orthogonal), localized cost functions, and use of shallow, task-specific ansätze [21].

III. Literature review on QML4EO

A. Review Methodology

To ensure a comprehensive and unbiased review, we employed a structured methodology to collect relevant works on QML applied to EO. Our approach was designed to maximize coverage while maintaining high standards of

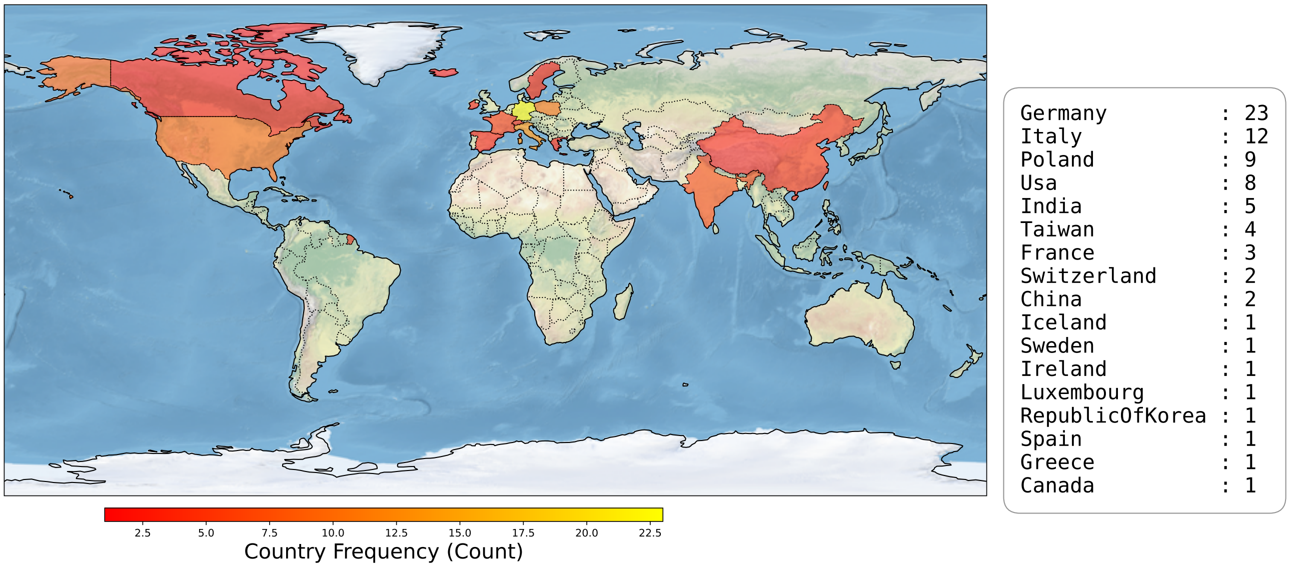

Figure 3. Geographical distribution of the contributions identified in our study, based on the affiliation of the first author. The map highlights a strong European presence, with Germany, Italy, and Poland leading in the number of contributions. The right-hand box provides a detailed numerical breakdown by country, illustrating the global interest in this emerging research area.

relevance and quality. The key steps of our methodology are outlined as follows.

We conducted systematic searches across multiple prominent scientific databases, including Web of Science, Scopus, IEEE Xplore, and arXiv. To enhance the completeness of our search, we formulate queries incorporating a combination of key terms related to QML and EO. Boolean operators and wildcards were used to refine search results and optimize the balance between precision and recall. An example of a search query used in our process is as follows:

("Quantum Machine Learning" OR "Quantum Neural Networks") AND ("Earth Observation" OR "Remote Sensing" OR "Satellite Data").

We iteratively refined our search terms based on initial retrievals and we applied a set of inclusion and exclusion criteria. Specifically, we included:

- Peer-reviewed journal articles that provide substantial contributions to QML in EO.
- Conference papers that introduce novel methodologies, frameworks, or case studies with sufficient technical depth.
- Preprints available on arXiv, recognizing that many cutting-edge studies in this rapidly evolving domain are yet to be formally published.

Conversely, we excluded:

- Poster presentations and unpublished manuscripts hosted solely on personal webpages or non-reputable repositories.
- Articles lacking sufficient technical depth, such as those providing only high-level discussions without quantitative results.
- Works that, despite mentioning QML and EO, did not present a clear methodological application or technical contributions within the domain.

To enhance transparency and reproducibility, we have compiled a comprehensive list of all works included in this review. This list is publicly accessible through our dedicated webpage, awesome-qc4eo [2], where readers can explore the complete bibliography along with associated metadata. The webpage is hosted on GitHub, facilitating community engagement and fostering contributions from researchers and practitioners interested in expanding this evolving repository. By providing an open and interactive platform, we aim to encourage collaboration and the continuous enrichment of resources in the field of QML4EO.

The outcome of our research led to the identification of 76 relevant publications. Considering the relatively early stage of this research area, this represents a significant milestone in the field. To gain further insight into the global landscape, we analyzed the geographical distribution of contributions based on the affiliation of the first author, shown in Figure 3.

The analysis reveals a strong European presence, with Germany leading by a substantial margin (23 publications), followed by Italy (12) and Poland (9). Contributions from outside Europe include the United States (8), India (5), and Taiwan (4), indicating growing interest in the topic across multiple continents. Other countries with notable contributions include France (3), China and Switzerland (2), and several others with single entries,

[2]Awesome QC4EO GitHub page: https://alessandrosebastianelli.github.io/awesome-QC4EO/

such as Canada, the Republic of Korea, Spain, and Greece. The full breakdown is shown in the right box in Figure 3.

Editorial Note: We collected and categorized papers until June 15, 2025, once publicly available. We are open and happy to include missing references; this will be taken into account after this initial submission.

B. Collected Contributions

Figure 4 provides a comprehensive overview of quantum computing libraries and platforms, categorized by their compatibility with quantum hardware, implementation status (simulated or deployed), and degree of quantum-classical hybridization. The visualization maps major frameworks such as D-Wave Ocean, PennyLane, Qiskit, and TorchQuantum, indicating whether they are hybrid, fully quantum, or quantum-inspired, and whether they are tied to specific hardware backends like IBM, D-Wave, or Rigetti. Notably, the distribution reveals a strong bias toward platforms associated with unknown or unspecified quantum hardware, suggesting that most contributions are currently focused more on algorithmic development and simulation rather than direct hardware deployment. This highlights both the exploratory nature of the field and the current limitations in hardware accessibility, guiding researchers toward tools aligned with their computational goals.

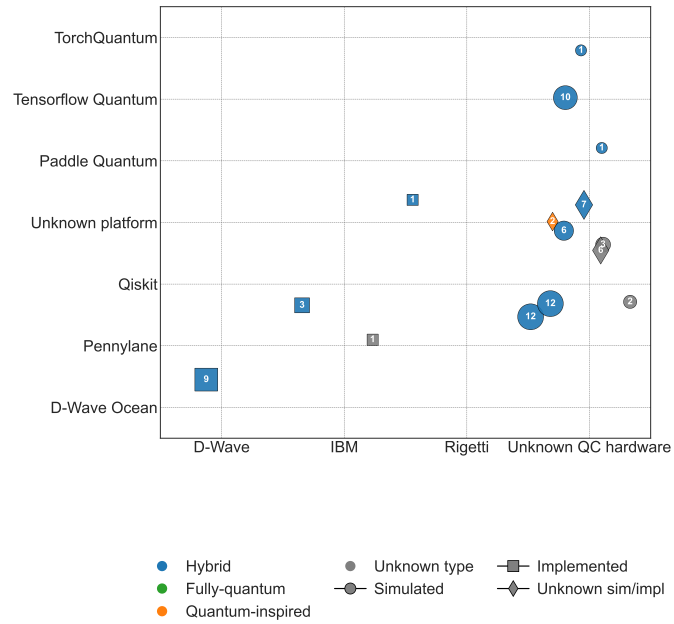

Figure 4. Classification of quantum computing libraries and platforms based on their quantum-classical hybrid nature, implementation status (simulated or deployed), and associated hardware (e.g., IBM, D-Wave, Rigetti). The figure also reveals a strong bias toward platforms with unknown or unspecified quantum hardware, indicating a current emphasis on algorithmic development over hardware-specific implementation. (Please note that jitter has been applied to (x,y) scatter coordinates to better separate overlapping scatterers).

Building on this landscape-level categorization, we now delve into the collected list of contributions according to the underlying QML model. This finer-grained analysis allows us to assess how different quantum machine learning approaches are employed across various platforms and research efforts.

In order to introduce the identified methodologies, better detailed in the next sections, we show in Fig. 5 summary graphic showing their configuration.

1) Quantum Neural Networks: Parametrized Quantum Circuits (QNNs) are quantum algorithms that leverage QC to learn and extract features from input data for analysis.

Hybrid Quantum Neural Network This type of model combines quantum and classical computing components to perform machine learning tasks. Depending on the role and placement of the quantum component in the architecture, these hybrid models can be further broadly classified into two categories, here called Early Hybrid Schemes and Late Hybrid Schemes.

- *Early Hybrid Schemes*: These models mainly exploit quantum computing to process input data through iterative encoding and analysis using QML techniques. A representative example is the quanvolutional neural network [22], which applies a PQC to small regions of the input to extract local features for subsequent analysis. By systematically sliding the PQC across the input, this approach enables large-scale EO image processing. The resulting quantum-processed features are then fed into classical models for downstream tasks, offering a promising strategy for integrating QML into EO data analysis during the NISQ era.

Based on these principles, various models have been proposed and have demonstrated strong potential for EO data analysis. For example, Henderson et al. [23] and Ghosh et al. [24] developed specialized PQCs to extract enhanced features from multispectral imagery and radar sounder data, respectively. In the context of Synthetic Aperture Radar (SAR) data, Miller et al. [25] employed quanvolution to generate feature maps for classification tasks, while Mauro et al. [26] explored its potential for speckle noise reduction. Furthermore, Sebastianelli et al. [27] introduced a quanvolution-based framework utilizing random circuits to process large volumes of diverse EO datasets, and proposed a parallelized quanvolutional operation to improve computational efficiency. Building on the findings reported in [27], Mauro et al. [28] enhanced a framework for estimating turbidity levels with data from the ΦSat-2 mission, by applying quanvolutional filters, as detailed in [29]. Departing from the conventional approach, Chalumuri et al. [30] developed a row-wise quanvolutional operation to extract local features for EO data classification. Otgonbaatar et al. [18] applied a PQC for pixel-wise classification of PolSAR imagery, and Lin et al. [31] integrated pixel-level features derived from QML into a Graph Neural Network (GNN) architecture for the hyperspectral image change detection task. In contrast, some studies [32]–[36] have explored full-input quantum encoding for feature extraction. However, these approaches face

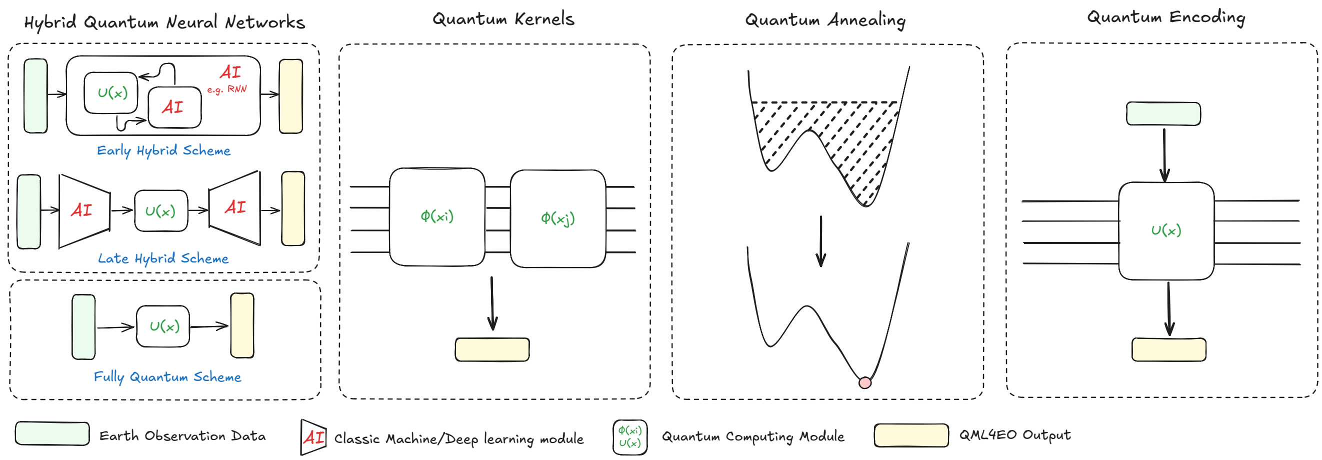

Figure 5. Identified quantum machine learning methodologies in the field of earth observation. From left to right: 1) Hybrid quantum neural networks, divided into A) early hybrid scheme, were QC is added internally to an AI module; B) late hybrid scheme, were QC is applied to AI-reduce set of feature; C) fully quantum scheme, were no classical AI is involved, which represent the aimed solution in the future; 2) Quantum Kernels; 3) Quantum Annealing; 4) Quantum Encoding Methods.

significant practical limitations due to the constraints of current NISQ hardware. Current implementations typically require input downsampling to fit within available quantum resources, which can lead to the loss of critical spatial-spectral information. This fundamental trade-off poses a significant challenge, particularly in EO applications where the preservation of fine-grained details could be crucial for effective analysis. Furthermore, this principle has also been applied to explore the potential of QML for fusing multiple EO modalities in data analysis. For instance, Majji et al. [37] and Miller et al. [38] employed different PQCs to integrate SAR and optical imagery, with the fused features subsequently processed by classical deep learning models for classification.

Park et al. [39] propose a novel hybrid quantum federated learning framework tailored for satellite-ground communication. Leveraging slimmable quantum federated learning (SQFL) and slimmable quantum neural networks (sQNNs) with angle and pole configurations, their approach enhances communication and computing efficiency. Techniques like superposition coding and successive decoding further increase communication opportunities. Their extensive experiments demonstrate that the satellite-ground SQFL framework outperforms classical federated learning and conventional QFL methods in both computational and communicational aspects, marking a promising advance in distributed quantum machine learning for EO applications.

- *Late Hybrid Schemes*: These hybrid models employ classical machine learning techniques to extract global feature representations, which are then processed by QML components to enhance higher-level feature refinement for final predictions. As a result, the selection and design of classical feature reduction methods are critical to the overall performance of such hybrid architectures, particularly under the constraints of NISQ hardware.

To date, various classical algorithms have been integrated with PQCs for EO tasks. Principal Component Analysis (PCA) is one typical algorithm that linearly transforms input data into a set of principal components. Gawron et al. [40] demonstrated the effectiveness of this framework for binary classification of multispectral images. Similar strategies have been extended to hyperspectral imagery: Priyanka et al. [41] implemented a QNN, and Shaik et al. [42] employed a Quantum Support Vector Machine (QSVM) for pixel-wise classification, both using PCA-based feature reduction.

In addition, deep learning, aiming to learn deep representations of data, has been extensively studied and has led to significant breakthroughs over the past decade, as reviewed in [43]. Utilizing such techniques in QML and combining them with QML has triggered great attention in research. For instance, Liliopoulos et al. [44] concentrated on Multilayer Perceptron (MLP)-based models that integrate quantum neurons in the hidden classical layers and achieved enhanced performance. Rainjonneau et al. [45] combined MLPs with a PQC as a quantum layer for satellite mission planning. Fan et al. [46] exploited MLP to assist EO data encoding for efficient QML analysis.

Besides traditional feedforward neural networks, Convolutional Neural Networks (CNNs) have proven particularly effective for EO imagery processing due to their ability to exploit spatial locality and perform hierarchical feature learning for efficient and robust feature extraction. Accordingly, many hybrid models, such as [47]–[49], incorporate CNN-based architectures to extract a set of informative features from EO data, and these features are then further processed using PQCs to perform the final classification with fully connected layers. In another study [50], Malarvanan et al. divided the features extracted by a CNN into four segments, each processed by a separate PQC. The outputs of these PQCs were then concatenated and passed through a fully connected layer for classification. Besides directly applying CNNs for feature reduction before feeding the data into PQCs, Zollner et al. [51] leveraged a deep convolutional network (VGG16) prior to the QML component for classification, and Chang et al. [52] utilized a convolutional autoencoder in their proposed model to reduce the input data dimensionality. The compressed feature representations were subsequently processed by a PQC for binary classification. In a study by Zollner et al. [53], several dimensionality reduction techniques were compared, and the results demonstrated the superiority of the autoencoder models in generating compact representations of EO imagery in comparison to traditional methods such as PCA. Otgonbaatar et al. [54] combined a convolutional autoencoder with a VGG16 to extract global features from multispectral EO images, which were then processed by a PQC for further classification. Such autoencoder-based hybrid models have been applied not only to classification tasks but also to EO data generation-related applications. For example, [55], [56] proposed to use QML for latent space transformations in hyperspectral image restoration. Ghosh et al. [57] introduced a hybrid model with a UNet-like architecture using a PQC to process the bottleneck features for semantic segmentation of radar sounder data.

Moreover, GNN architectures and attention mechanism have also gained popularity in EO applications, and their integration with QML has also been explored. For example, Mauro et al. [58] proposed combining a GNN with QML to predict the Ocean Niño Index. Papa et al. [59] examined the validity of integrating QML into ViTs for EO applications.

To better tackle the challenges in EO data analysis using DL models, various comprehensive training strategies have been explored and widely adopted in practice. These strategies have also been integrated into hybrid QML models. One representative example is the use of pretrained models, which aim to enhance performance and generalization under limited annotated data via transfer learning. A study by Otgonbaatar et al. [60] demonstrated the effectiveness of this framework when applied to a small but high-dimensional annotated EO dataset. Chang et al. [61] used a pretrained autoencoder to map high-dimensional data into a latent space and proposed a hybrid classical-quantum GAN for an image generation task. In a related study, De Falco et al. [62] also exploited QML within the latent space and developed a quantum latent diffusion model integrated with a convolutional autoencoder for EO imagery generation. Besides that, ensemble learning has also gained attention in the development of hybrid models. For instance, Bhavsar et al. [63] employed multiple classical machine learning algorithms for robust feature reduction, followed by a PQC for prediction.

Van et al. [64] compare three distinct quantum neural network architectures for multispectral image classification using the EuroSAT dataset. Their study includes a fully quantum convolutional neural network (QCNN), as well as hybrid and orthogonal variants, implemented via TensorFlow Quantum. The results highlight that both hybrid and fully quantum approaches offer improvements in classification performance, suggesting a growing relevance of QNNs for remote sensing tasks as quantum computing matures.

Glatting et al. [65] propose a cognitive SAR architecture for ship detection based on quantum reinforcement learning. The system utilizes two satellites in coordination: one with wide-swath, low-resolution scanning and another with high-resolution beam refocusing. Quantum variational circuits are employed within the reinforcement learning framework to dynamically adapt sensing strategies, highlighting the potential of QML for intelligent decision-making in EO scenarios.

A recent study by Otgonbaatar et al. [66] emphasizes that a true quantum advantage can only be expected if quantum models are able to generalize better than classical ones and avoid limitations such as weight symmetry during training. The authors highlight hyperspectral image classification as a particularly promising application domain due to its smaller input sizes and limited availability of training data, although current hardware constraints remain a bottleneck.

These studies collectively highlight the paradigm for feasibly processing EO data using quantum computing in the NISQ era. However, whether QNNs can consistently outperform their classical counterparts remains an open question and requires further investigation.

Fully Quantum Neural Network This type of model employs quantum computing exclusively for all stages of the learning process, including data encoding, feature extraction and prediction. Unlike hybrid approaches, it does not incorporate any classical components for processing and analyzing data.

Due to the limitations of quantum resources in the NISQ area, there are relatively few studies employing such models for analyzing EO data, which typically contain rich spatial and spectral information. Nevertheless, notable efforts have been made to explore the potential of QML in the EO domain under current constraints of quantum resources. For instance, Painchart et al. [67] introduced a QNN with a pyramid structure to reduce quantum resources for analyzing and classifying temporal sequence EO data. Yu et al. [68] introduced an evolutionary algorithm-driven method for quantum circuit design, aimed at minimizing quantum resource usage within an image similarity learning framework for effective representation learning. Pastori et al. [69] explore the use of QNNs to model cloud cover parameterizations in climate models, finding out that QNNs can capture non-linear relationships relevant to cloud dynamics and demonstrating potential for parameter reduction and improved generalization in constrained settings.

2) Quantum Kernels: Recent research has increasingly explored quantum kernel methods as a promising approach to EO tasks, particularly in settings where classical kernel-based models like support vector machines already perform well. In this context, quantum kernels refer to similarity measures computed using quantum circuits, where classical data is embedded into a quantum Hilbert space and inner products are estimated via quantum measurements. These kernels can, in theory, capture highly nonlinear structures in data using fewer resources than their classical counterparts, depending on the nature of the embedding.

Miroszewski et al. [7] provide a broad overview of QML applications in Remote Sensing (RS), identifying both the potential of quantum kernels and the limitations posed by current hardware. A series of focused studies by the same group demonstrates practical implementations, notably in cloud detection using multispectral Sentinel-2 imagery [70], [71] and semantic segmentation using hyperspectral sensors [72]. Their results show that quantum kernel SVMs can match or slightly outperform classical counterparts on small-scale datasets, offering benefits in generalization and feature mapping. Shaik et al. [73] test simulated quantum kernels on a feature-reduced hyperspectral dataset, obtaining a 5% improvement in accuracy over RBF kernels for 12 features. Shaik et al. [42] extend the utility of quantum kernel methods into the semi-supervised learning regime, proposing a quantum-based pseudo-labelling approach for hyperspectral image classification that improves performance when labeled data is scarce. Sarkar et al. [74] apply quantum-enhanced SVMs to multiclass land use / land cover classification, reporting encouraging results that point toward hybrid workflows as a practical direction. Glatting et al. [65] introduce quantum kernels for InSAR phase unwrapping, a novel application where phase continuity and optimization complexity make quantum approaches conceptually attractive. Meanwhile, Rodriguez-Grasa et al. [75] explore neural quantum kernels for satellite image classification, proposing a hybrid architecture that combines classical neural networks with quantum feature maps. Theoretical and practical considerations are deepened by Miroszewski et al. [76], who analyze the impact of shot count on kernel estimation, highlighting the trade-offs between quantum fidelity and computational noise. Across these studies, a consistent theme emerges: quantum kernels are not yet outperforming classical baselines across the board. Nevertheless, they offer competitive accuracy, greater flexibility in non-linear mapping, and a testbed for exploring quantum advantage in real EO problems [77], [78].

3) Quantum Data Encoding: Both QNNs and quantum kernels require classical data to be converted to the quantum domain, in a step called quantum data encoding. The choice of encoding affects not only the expressiveness and efficiency of the quantum model but also its compatibility with hardware constraints and the nature of the input data. Some works have put strong emphasis on this step, finding specific approaches for EO data.

Zollner et al. [53] explore different satellite image representation strategies for quantum classifiers, comparing encoding schemes such as amplitude encoding, angle encoding, and hybrid variants in terms of dimensionality, noise resilience, and classification accuracy. Their findings emphasize the trade-offs between circuit depth, interpretability, and performance, particularly in the context of image data with strong spatial structure. In a more specialized application, Otgonbaatar and Datcu [18] propose a natural quantum embedding of Stokes parameters derived from polarimetric SAR imagery. By directly mapping the physical quantities of polarimetric data into quantum gates, their method aligns with the physics of both the sensor and the quantum system, preserving important structural and statistical properties. Together, these works highlight the emerging focus on encoding as a design challenge, especially when adapting EO data, which often has high dimensionality and complex modality, to the constraints and potential of quantum models.

4) Quantum Annealing: Quantum Annealing (QA) is a metaheuristic optimization technique that finds low-energy configurations in an energy landscape. The intrinsic suitability of QA for combinatorial optimization makes it appealing for EO problems, which often involve high-dimensional spaces, non-convex optimization, and discrete decision variables. Quantum annealers developed by D-Wave Systems are already being used to address real-world EO tasks.

Classification of remote sensing data remains one of the most explored areas for QA applications, where optimization tasks appear. Boyda et al. [79] demonstrated an early proof-of-concept by detecting tree cover in aerial images using QA. Though limited by hardware constraints at the time, their study paved the way for later work on QA-enabled environmental monitoring. Otgonbaatar and Datcu [80] proposed a QA-based feature subset selection method for hyperspectral image classification. Their approach enhances classification accuracy by focusing on the most informative spectral bands, which also reduces the Quadratic Unconstrained Binary Optimization (QUBO) size and improves quantum resource efficiency. Delilbasic et al. [81] developed a quantum-trained SVM method tailored to binary EO datasets, leveraging QA to solve the QUBO form of SVMs. Their results marked a significant step toward quantum-enhanced EO data classification. In later work, Delilbasic et al. [82] extended the quantum-trained SVM method to multiclass classification. This approach formulates the entire multiclass classification problem as a single optimization task, avoiding one-vs-all decomposition. Their method demonstrates competitive accuracy compared to classical approaches while reducing computational overhead. Then, Zardini et al. [83] proposed a framework for adding locality to quantum-trained binary and multiclass SVMs, demonstrating how

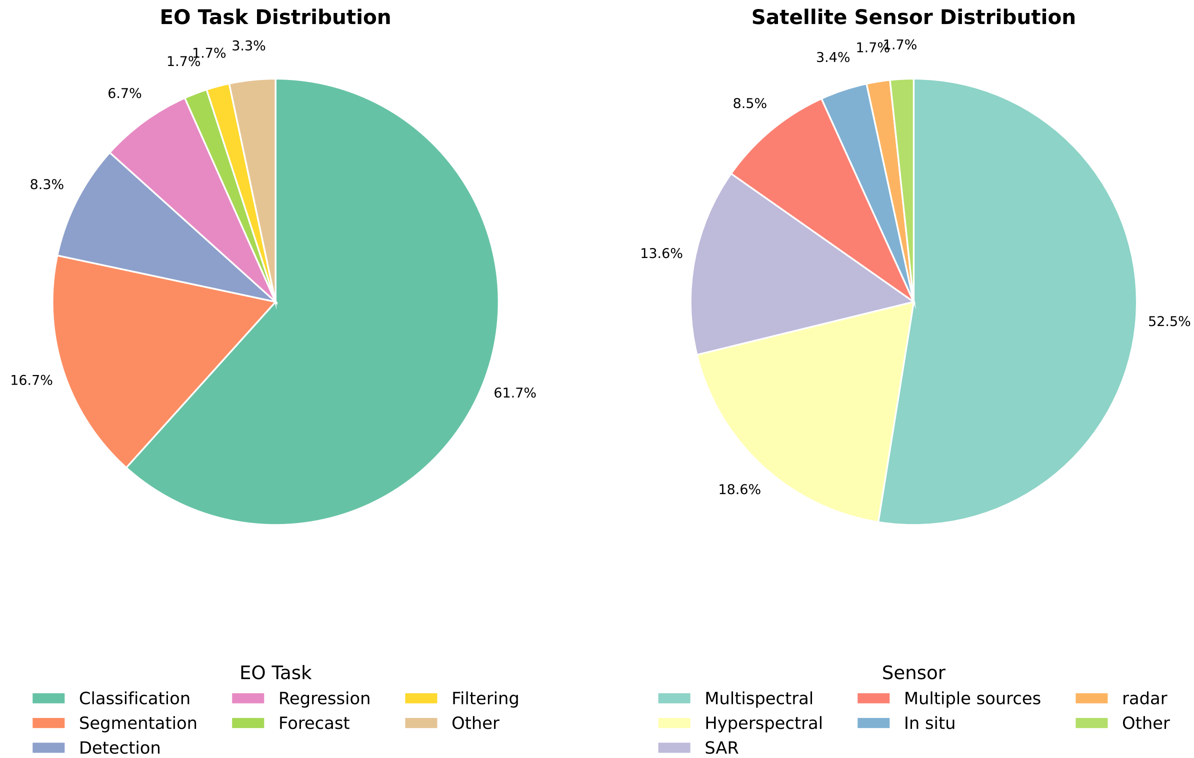

Figure 6. Distribution of QML applications across EO sensor modalities (left) and task types (right).

restricting the training set to the k nearest neighbors can improve scalability and robustness under current hardware constraints. Delilbasic et al. [84] summarized advantages and limitations of SVM models trained with quantum annealing.

Regression tasks in EO, such as estimating biophysical variables, present unique challenges due to the continuous nature of outputs. Pasetto et al. [85] proposed a QA-based support vector regression model for estimating variables like leaf area index. Their implementation translates regression objectives into QUBO formulations, enabling deployment on existing quantum hardware. Pasetto et al. [86] also investigated kernel approximation techniques on a quantum annealer for EO regression tasks. By approximating non-linear kernels efficiently within the QUBO framework, the study highlights new avenues for leveraging QA in continuous prediction scenarios. Efficient data handling is critical for EO, especially with high-dimensional inputs. Otgonbaatar and Datcu [87] also tackled the coreset selection problem, aiming to build smaller yet informative datasets for downstream tasks. Their method leverages classical algorithms to select representative subsets of images, balancing fidelity with computational feasibility. They then train a QSVM using these subsets.

5) Quantum-inspired Methods: For completeness, we also include a few works based on quantum-inspired methods, i.e., classical algorithms or architectures that draw conceptual or mathematical inspiration from quantum mechanics, but are designed for conventional hardware. These approaches often aim to capture the advantages of quantum formalisms without the overhead and noise of real quantum devices. One typical model is the tensor network, an important concept in many-body quantum systems that provides an efficient representation of quantum states [88]. Matrix Product States (MPS) is one of the fundamental examples, widely used to represent entangled states in low-dimensional quantum systems. Inspired by this, Otgonbaatar et al. [89] applied tensor network techniques to reduce the number of parameters in a deep learning model and improve the spectral resolution of Hyperspectral images (HSIs) by decomposing them into sparse factor matrices. In another example, motivated by the quantum theory, Zhang et al. [90] proposed a spectral-spatial network for HSI feature extraction. The network includes a phase-prediction module that assigns a phase for each input feature based on discriminative information, and a measurement-like fusion module that generates a low-dimensional feature representation for further analysis.

IV. Application Domains and Research Directions in QML4EO

QML has emerged as a promising avenue for tackling the computational and data complexity challenges inherent in modern EO, particularly in processing high-dimensional, multi-source remote sensing datasets. This section synthesizes the current landscape of QML applications in EO, highlighting prevalent data types, methodological trends, task categories, and institutional actors shaping the field.

Table II
Representative QML applications in Earth Observation (EO).

Application Area	EO Data Type	Quantum Method	Reference
Land cover classification	Multispectral	VQC + CNN	Sebastianelli et al. [48]
Deforestation detection (time series)	SAR (Sentinel-1)	Quantum CNN + RNN	Painchart et al. [67]
Cloud detection	Multispectral	Quantum kernel	Miroszewski et al. [70]
EO classification (EuroSAT, Sentinel-1)	Multispectral + SAR	Quanvolutional neural network (QuanvNN)	Sebastianelli et al. [17]
Semantic segmentation	Multispectral	Quantum annealing	Delilbasic et al. [84]
Multisource fusion for classification	SAR + Optical	Quantum-enhanced preprocessing	Majji et al. [37]
Feature learning	Multispectral	Quantum contrastive similarity learning	Yu et al. [68]
Hyperspectral compression	Hyperspectral	Tensor networks	Otgonbaatar et al. [89]

A. Emerging Patterns in EO Sensors and Tasks

Our analysis reveals a concentration of QML applications around specific sensor modalities and EO task types, as illustrated in Figure 6.

More than 50% of the studies leverage multispectral imagery, followed by hyperspectral at 18.6% and SAR data at 13.6%. A smaller subset of works explores multi-sensor fusion or incorporates in-situ measurements.

Classification tasks dominate the field, accounting for nearly 62% of QML applications—these are further detailed in Section IV-B1. Segmentation (16.7%) and object detection (8.3%) follow, commonly applied to urban mapping and ship detection. Regression, forecasting, and filtering remain underexplored but offer fertile ground for future developments, particularly in climate modeling and environmental monitoring.

B. Common Quantum Approaches and Technological Trends

Most QML models for EO rely on hybrid quantum-classical architectures tailored to the constraints of NISQ hardware. At the core of these systems are Variational Quantum Circuit (VQC)s, which enable low-depth, parameter-efficient learning via classical optimization.

Key trends include:

- VQCs paired with classical CNNs for land cover classification [48];
- Quantum kernel methods for hyperspectral pixel classification and cloud detection [70], [75];
- Quanvolutional layers integrated into classical deep networks for enhanced feature extraction from multispectral and SAR data [17], [52];
- Quantum-inspired tensor networks for dimensionality reduction in hyperspectral data cubes [89];

While classification is the dominant focus, a small but growing number of studies extend QML methods to structured prediction, regression, and data fusion tasks. A summary of representative QML4EO applications, including the data modalities and quantum methods used, is presented in Table II.

1) Land Cover and Crop Type Classification: Land cover and crop type classification represent the most active areas of QML application in EO, often using multispectral or hyperspectral inputs.

VQCs have shown strong potential in these tasks, offering comparable performance to classical CNNs with significantly fewer parameters. Sebastianelli et al. [48], for example, matched CNN accuracy on the EuroSAT dataset using a compact hybrid quantum-classical model.

In parallel, quantum kernel methods have gained traction for their ability to embed input data into high-dimensional Hilbert spaces. Rodriguez et al. [75] introduced a neural quantum kernel classifier that outperformed classical SVM baselines in multispectral classification, demonstrating the value of quantum-aware feature encodings.

More recently, Sebastianelli et al. [17] proposed the Quanv4EO model, which incorporates quanvolutional layers into the preprocessing pipeline of hybrid quantum-classical architectures. Applied to multisource datasets such as EuroSAT and Sentinel-1, this model demonstrated improved classification accuracy (up to 5% gain) and 80% faster convergence during training compared to classical baselines, along with notable parameter efficiency.

2) Semantic Segmentation: Quantum-enhanced models are also emerging in structured output tasks such as segmentation and detection, where spatial context and fine-grained boundaries are essential.

Delilbasic et al. [84] investigated the use of quantum annealing for semantic segmentation using multispectral EO imagery. Their study highlighted both the potential and limitations of current quantum annealing hardware, emphasizing the importance of problem encoding and postprocessing for effective label assignment.

3) Geophysical Parameter Estimation: QML has also been applied to continuous geophysical variable retrieval from satellite data—an inherently nonlinear inverse problem.

Painchart et al. [67] introduced a quantum neural network to detect deforestation from temporal SAR sequences. Although framed as a classification task, their model demonstrated the feasibility of applying quantum convolutional and recurrent layers to extract dynamic features from time-series EO data.

4) Representation Learning for Feature Similarity: Recent studies have explored QML's potential for learning efficient and transferable data representations.

Yu et al. [68] proposed a quantum contrastive learning approach based on unsupervised similarity triplets. Applied to multispectral imagery, the model achieved significant reduction in quantum resource usage while maintaining strong performance in downstream tasks such as land surface classification and deforestation detection.

5) Multimodal Data Fusion and Dimensionality Reduction: As EO data becomes increasingly multimodal, QML offers novel strategies for data fusion and compression.

Majji et al. [37] proposed a quantum-inspired framework for enhancing SAR and optical image fusion. By applying a set of quantum image processing techniques, including quantum Fourier transform and Hadamard-based fusion operators, the authors demonstrated improved classification accuracy in land-use tasks, with performance gains over SAR-only inputs and standard fusion methods. Though the processing was simulated, the study illustrates the potential of quantum-inspired operations to improve EO data interpretability.

In parallel, Otgonbaatar et al. [89] employed tensor network decomposition techniques, originally developed in quantum many-body physics, for hyperspectral data compression. The approach achieved competitive classification accuracy while reducing input dimensionality, offering an efficient pathway for spectral feature extraction in high-dimensional EO datasets.

6) Generative Models and Data Augmentation: Quantum generative models are still nascent but show promise for EO data synthesis and augmentation.

Zhang et al. [90] developed a Quantum GAN (QGAN) to generate realistic hyperspectral image patches. Used for data augmentation, the synthetic samples improved downstream classification accuracy by 3–4%, underscoring their utility in low-sample regimes.

C. Emerging Use Cases and Research Directions

Beyond pattern recognition, recent studies point to broader QML applications in EO, including:

- **Time-series modeling** with quantum recurrent architectures for change detection and climate dynamics;
- **Probabilistic inversion** using quantum-enhanced Bayesian inference for biophysical parameter retrieval;
- **Onboard processing** via lightweight quantum circuits for low-latency satellite edge computing.

These emerging directions highlight QML's potential to advance EO well beyond classification and segmentation, into dynamic, probabilistic, and real-time domains.

D. Geographic and Institutional Ecosystem

The field of QML4EO is supported by a growing and geographically diverse research community. European research centers such as ESA's Φ-lab, the German Aerospace Center (DLR), and the Technical University of Munich (TUM) have been particularly active, producing multiple foundational studies and benchmarking frameworks. Other notable contributors include:

- The Jülich Supercomputing Centre and University of Iceland, with contributions in quantum optimization and EO image segmentation;
- NASA-affiliated researchers and U.S. universities exploring quantum annealing techniques for mission planning and resource allocation;
- Asian academic institutions, with a focus on quantum-inspired approaches to hyperspectral image analysis.
- University of Sannio and University of Pavia, with a focus on hybrid quantum neural networks applied to multispectral and SAR data

A significant step toward structural consolidation has been the establishment of the *Quantum Earth Science and Technology (QUEST)* Technical Committee within the IEEE Geoscience and Remote Sensing Society (GRSS). QUEST plays a pivotal role in shaping the research agenda by promoting open-source software tools, organizing community benchmarks, and fostering interdisciplinary dialogue between EO scientists and quantum computing researchers.

V. Discussion

The literature on QML4EO has grown exponentially since 2017, yet its centre of gravity remains narrow and preliminary. Figure 4 of our review shows that about 73% of the papers we analysed executed their quantum circuits only on classical simulators, and a further 15% combined simulation with a single "proof-of-concept" hardware run involving < 10 qubits. Only 4% relied exclusively on physical devices and none of these exceeded 33 qubits. This heavy tilt towards simulated back-ends explains why reported accuracies sometimes match or slightly exceed classical CNN or SVM baselines: the experiments are effectively noise-free, unconstrained by qubit topology, and run with unlimited shot counts.

A second trend is the over-concentration on pixel-wise or patch-wise land-cover classification (60% of studies) and SAR/hyperspectral variants thereof. Tasks that arguably showcase the distinctive strengths of quantum kernels, e.g. data fusion across modalities, probabilistic inversion, or large-scale spatio-temporal forecasting, account for $< 10\%$ of the corpus. This thematic imbalance limits the external validity of today's "quantum advantage" claims: if the community keeps benchmarking on simplified, low-resolution or heavily down-sampled datasets, it will remain unclear whether any reported speed-ups survive contact with real EO workloads.

A. Methodological Weak Spots

Classical baselines are rarely optimised to parity. Several "quantum-wins" disappear when the comparator model is given the same parameter budget or when standard regularisation is applied. Worse, inference latency and energy for the cryogenic stack are never included in cost metrics, creating a distorted "qubit-seconds per prediction" narrative.

Only 48% of the papers we surveyed publish code or circuit specifications. Different cloud providers still lack a common intermediate representation, making cross-platform replication difficult. Without transparent kernels and random seeds, it is impossible to separate algorithmic merit from hyper-parameter luck.

B. Hardware Reality Check

In late 2023, IBM announced the 1121-qubit "Condor" chip, and in 2025 a roadmap toward the 1386-qubit multi-chip "Kookaburra" [91]. These numbers sound impressive until one realises that (i) usable qubit counts after error-mitigation are one to two orders of magnitude lower, and (ii) single- and two-qubit gate fidelities remain in the 99% − 99.5% band, meaning error probabilities compound rapidly with circuit depth. Coherence times on today's superconducting platforms are still in the order of $100\mu s - 1ms$, constraining practical circuits to a few hundred layers at best [92]. Consequently, most QML4EO prototypes cap circuit depth at < 20 parameterised gates, a regime where classical tensor-network simulators can still track the state vector efficiently.

C. Where Does This Leave "Quantum Advantage"?

The evidence collected in this review suggests that an unambiguous, end-to-end quantum speed-up for operational EO pipelines is not yet within reach; however, the data also reveal encouraging trends that deserve equal attention. First, hardware scaling is beginning to intersect with EO-relevant problem sizes. When the field launched in 2017, public devices exposed fewer than 20 high-fidelity qubits; today laboratory prototypes already pass the one-thousand-qubit mark, and cloud back-ends with 100–200 usable qubits are routinely available. Although error-rates and qubit connectivity still limit circuit depth, the ability to embed tens of independent EO features without extreme dimensionality reduction is now realistic. This leap removes one of the most frequently cited "show-stoppers" and enables explorations that were previously confined to idealised simulators.

Second, hybrid quantum–classical training loops are maturing quickly. Gradient-free optimisers, layerwise learning-rate schedules and automatic shot-allocation have collectively contributed to the reduction of wall-clock training time [93]–[95]. These engineering improvements do not constitute "quantum advantage" in a strict complexity-theoretic sense, but they do narrow the practical gap between research prototypes and production-grade EO analytics, particularly for tasks where classical meta-learning struggles with vanishing gradients or prohibitive memory footprints.

Third, community practice is becoming more rigorous. More than half of the papers published after 2024 deposit circuit descriptions, random seeds and pre-processing scripts in public repositories, making independent replication possible for the first time. In parallel, several open-access benchmark suites (e.g., QEarthNet-B and Q4SAR-Lite) now specify fixed qubit budgets, shot limits and energy-usage reporting. These resources provide the uniform baselines that are essential for credible "advantage" claims and will help filter genuine algorithmic breakthroughs from artefacts of hyper-parameter tuning.

Taken together, these observations motivate a more nuanced position. QML4EO has transitioned from speculative curiosity to methodologically sound, steadily progressing research programme. Near-term "quantum advantage" will almost certainly be task-specific and resource-bounded, e.g., accelerated kernel evaluations in small-footprint edge constellations, or improved robustness against covariate shift in multi-modal fusion, rather than a sweeping replacement of classical deep learning. The path forward is therefore clear:

1) Target EO problems whose structure aligns with currently available qubit counts and noise profiles;
2) Report holistic metrics, including latency, energy and reproducibility, so that efficiency claims remain grounded; and
3) Invest in open benchmarks that couple real-world data complexity with transparent, hardware-aware constraints.

In that frame, the question "Where does this leave quantum advantage?" can be answered cautiously but optimistically: on the horizon, no longer on the drawing board. We do not yet have conclusive proof, but the convergence of maturing hardware, sharper algorithms and disciplined benchmarking suggests that domain-relevant advantage, though likely incremental rather than monumental, may emerge sooner than many sceptics predicted.

VI. Conclusion

QC presents a remarkable opportunity to improve EO by addressing the computational hurdles that currently limit its capabilities. While QML has shown considerable promise in EO applications, several limitations currently constrain its broader adoption:

- Most QML implementations are executed on classical quantum simulators, with real-device execution limited to few-qubit proof-of-concept studies;
- A predominant focus on classification tasks may obscure QML's unique value for inference-heavy problems such as geophysical parameter retrieval, uncertainty quantification, or spatiotemporal forecasting;
- The absence of standardized benchmarks and reproducibility protocols hinders rigorous comparison and evaluation across methods.

Despite these challenges, the field is rapidly evolving. As shown in Figure 6, the methodological base is expanding, and novel applications are beginning to emerge. Continued improvements in quantum hardware, enhanced algorithmic robustness, and the adoption of community-wide benchmarking practices are expected to drive QML from a niche research domain toward becoming a complementary component within operational EO pipelines.

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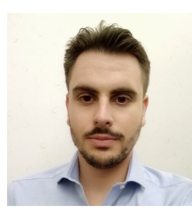

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